

Reg. No.:											
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Question Paper Code: 28111
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**SARANATHAN COLLEGE OF ENGINEERING, TIRUCHIRAPPALLI - 620 012
(AUTONOMOUS)**

B.E./B.Tech. END SEMESTER EXAMINATIONS, NOV/DEC 2024

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

V Semester

CCS334 - BIG DATA ANALYTICS

(Regulation 2021)

Time: Three Hours

Max. Marks: 100

Answer ALL Questions

PART - A

(10 x 2 = 20 Marks)

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| 1. | List out the key trends in big data. | C01 | L1 |
| 2. | Define unstructured data with an example. | C01 | L1 |
| 3. | Examine the key value pair formation | C02 | L2 |
| 4. | Describe Cassandra clients. | C02 | L1 |
| 5. | Differentiate between Hadoop and Map Reduce. | C03 | L2 |
| 6. | Define the term scaling out. | C03 | L1 |
| 7. | Differentiate Cassandra vs Hadoop. | C04 | L2 |
| 8. | Define the term Name node and Data node. | C04 | L1 |
| 9. | Depict the need for Apache pig. | C05 | L2 |
| 10. | Classify the features of Hive. | C05 | L2 |

PART - B**(5 x 16 = 80 Marks)**

11. (a) i) Identify the benefits and challenges of using big data in industries? (8) CO1 L3
- ii) Explain the key characteristics and structure of big data? (8) CO1 L3

(OR)

11. (b) i) What are the key differences between the types of data used in Business Intelligence and Big Data analytics? (8) CO1 L3
- ii) How can Crowd sourcing be used to collect and analyze large amounts of data? (8) CO1 L3

12. (a) i) List the Classification of NoSQL Databases and explain about columns-based database. (8) CO2 L3
- ii) Explain detail about graph databases with a real time example. (8) CO2 L3

(OR)

12. (b) i) How do aggregate data models differ from relational data models? (8) CO2 L3
- ii) Explain detail about Cassandra clients with an example. (8) CO2 L3

13. (a) Illustrate the working model of map-reduce framework with an example. (16) CO3 L3

(OR)

13. (b) Analyze the factors that affect the performance of Map Reduce jobs when using different job schedulers. (16) CO3 L3

14. (a) Explain detail about design of Hadoop distributed file system (HDFS) (16) CO4 L3

(OR)

14. (b) Explain detail about File based data structures. (16) CO4 L3

15. (a) Demonstrate about Hbase and Hbase clients in detail. (16) CO5 L4

(OR)

15. (b) Analyze in detail about HiveQL data manipulation, queries and data types. (16) C05 L4
